EV CHARGING · NETWORK INTRUSION DETECTION

# EVNet Sentinel

## A leakage-corrected reproduction and comparative study of intrusion detection on EV charging networks — with a closed-loop response

**Shardul Chogale** · **Mukta Varak** · **Shruti Chaurasiya** · **Neha Chavhan** | Guide: **Prof. Mahesh Khandke**
Department of Computer Engineering, Vidyalankar Institute of Technology, Mumbai, India

Watch the video

Code & results

**99.8%**
accuracy from **one timestamp column** alone — the published 98.4% rests on a leak

**0.558**
honest **macro-F1** with the leak removed (± 0.008, 3 seeds)

**≥ 99.8%**
**floods** named exactly (95% lower bound) — scans named only 34%

**20 / 20**
recorded attacks **contained** in the response replay; 0 actions on normal traffic

## 1 Problem

Every public EV charger (EVSE) is a networked computer. It reports to a **charging management system (CSMS)** over OCPP, usually across the site's shared wifi, and talks to the car over ISO 15118. Flooding a charger cuts its link to the CSMS: sessions cannot be authorised or billed.

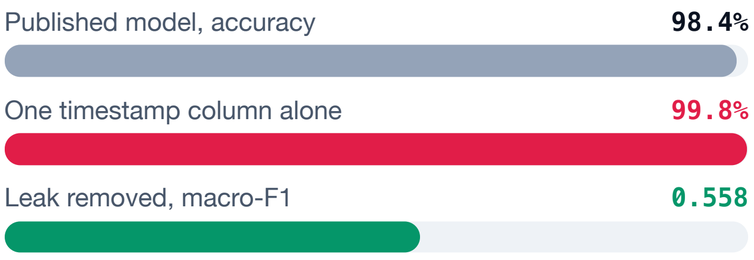
Makhmudov et al. [1] reported **98.4% accuracy** with an adaptive random forest and drift detection. We set out to reproduce it, and to go beyond detection.

## 2 Dataset

**CICEVSE2024** (Canadian Institute for Cybersecurity): traffic from two real charging stations, idle and charging, recorded from a mirrored switch port.

- **1,198,151** flows after de-duplication, **15** classes, **60** flow features
- Attacks: 5 volumetric floods, 6 nmap scan types, ICMP and Slowloris
- Senders: a laptop and a Raspberry Pi on the wifi, and a **compromised car** scanning over the charging cable

## 3 The leak



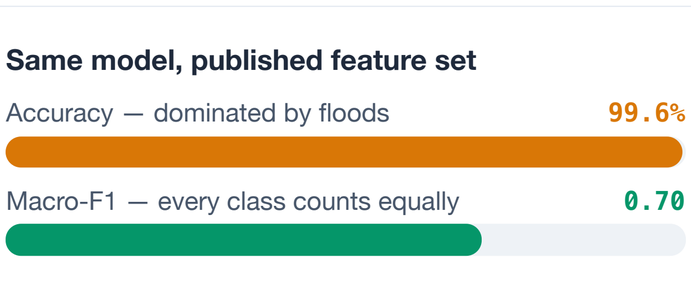

Six absolute capture timestamps identify the **recording**, not the traffic: a single column (`bidirectional_first_seen_ms`) out-scores the published model. The drift alarms also land on capture-file seams, not real concept drift.

## 4 Method

1. **Remove leakage** — drop the timestamps and `src_port`; de-duplicate flows.
2. **Split** — stratified train / validation / test, repeated for seeds 42, 1337, 2024.
3. **Detect** — six static models (XGBoost, LightGBM, Random Forest, Decision Tree, Logistic Regression, SVM) and an online ARF + ADWIN.
4. **Test per attack** — every class, every model, 30 × 50 unseen flows per model set; graded detected / right category / named.
5. **Respond** — a containment policy, replayed over every recorded attack (section 8).

## 5 Why macro-F1, not accuracy

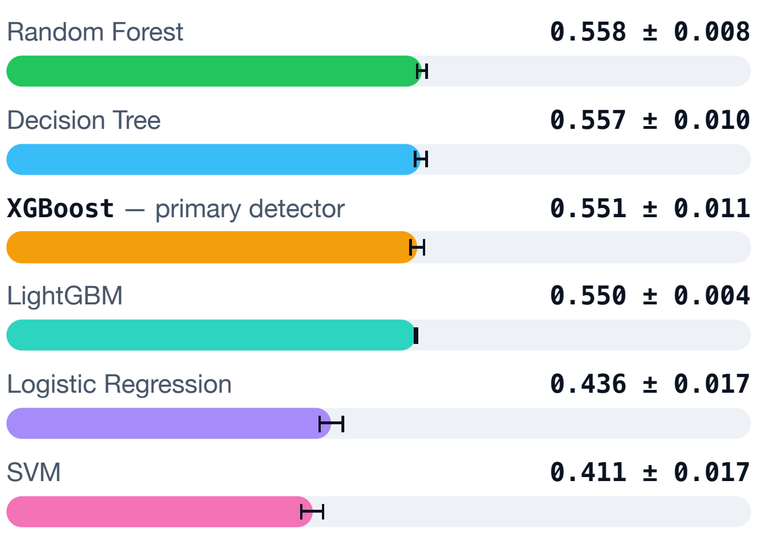
After de-duplication, floods are **over 80%** of the flows (99% on the published feature set). A model that names every flood and misses every scan still scores high accuracy.




Macro-F1 averages the F1 of each of the 15 classes, so a missed scan type weighs as much as a flood.

## 6 Corrected results

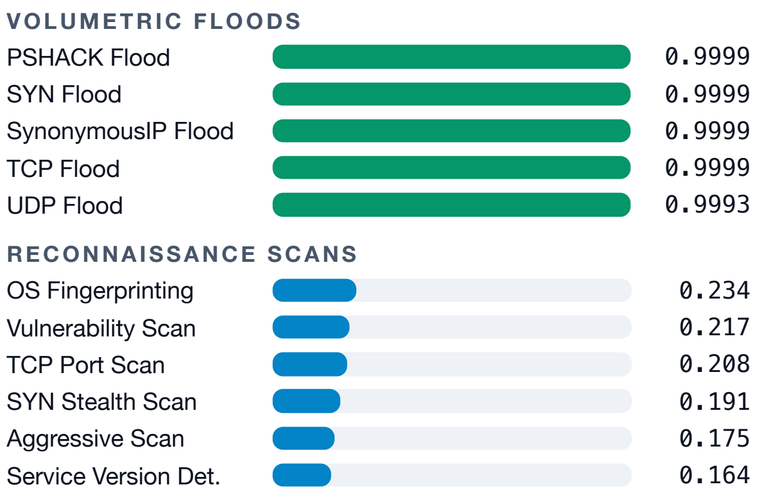
Macro-F1, mean ± std over 3 seeds (bar = mean, whisker = ± 1 std).




The four tree models tie within one standard deviation. **XGBoost** is the primary detector: the fewest false alarms on normal traffic, **1.8%** (Random Forest 2.6%).

## 7 Floods are solved. Scans are not.

Per-class F1, XGBoost, mean over 3 seeds.




The six nmap modes look almost identical at flow level. Still, **96%** of scans are recognised *as scans* — the response acts on that.

## 8 Closing the loop

Detection alone does not stop an attacker. Sentinel recommends an action matched to how reliable the detector is; **a person approves every action**, and the chargers and CSMS can never be cut off.

| Detect | Contain | Harden | Verify |
|---|---|---|---|
| classify each flow | one action per device | fix it for next time | attack flows stop |

| ATTACK | ACTION | WHO · HOW |
|---|---|---|
| Flood from the wifi | Quarantine | Analyst · switch port / ACL |
| Scan from the wifi | Restrict | Analyst · no access to chargers or CSMS; device watched |
| Scan from a compromised car | Stop session | CSMS operator · RemoteStopTransaction, block the vehicle's ID token |
| Watched device starts flooding | Quarantine | Escalation at a much lower bar |

## 9 Response, replayed

Every recorded attack, from the device that really sent it, streamed through the policy with XGBoost's real verdicts and an operator who approves after 5 flows.

**20/20**
attacker–attack pairs contained

**11 → 1**
flood flows to quarantine, cold vs after a scan

**0**
actions over 2,000 benign flows

A replay of recorded traffic through the policy, not live blocking: Sentinel is a passive, mirror-port IDS. The dataset has only 12 distinct benign flows, so the zero is a small sample.

## 10 Conclusions

- The published 98.4% comes from **capture timestamps**; corrected macro-F1 is **≈ 0.55** for every tree model.
- **Floods are solved** (≥ 99.8% named); **scans are data-limited**, not model-limited — stronger models do not lift the ceiling.
- A detector earns its keep in the **response**: category-level actions contain every recorded attack.

**Next:** session-level features to separate the scan types, and validation on a live OCPP testbed.

## Team

| | |
|---|---|
| **Shardul Chogale** | ML pipeline & evaluation |
| **Mukta Varak** | Cloud deployment & DevOps |
| **Neha Chavhan** | Frontend & API integration |
| **Shruti Chaurasiya** | Data visualisation & testing |

## 11 References

[1] F. Makhmudov, D. Kilichev, U. Giyosov, F. Akhmedov, "Online machine learning for intrusion detection in electric vehicle charging systems," *Mathematics*, 13(5):712, 2025.
[2] Canadian Institute for Cybersecurity, "CICEVSE2024 dataset," University of New Brunswick, 2024.
[3] J. Montiel et al., "River: Machine learning for streaming data in Python," *JMLR* 22(110), 2021.


Corresponding author: Shardul Chogale · shardulchogale1983@gmail.com